Combinatorial Analysis

Lecture 2

Topics in this lecture

- Basic counting principles
- Permutation
- Sampling

Basic Principle of Counting

- Suppose that 2 experiments are to be performed.
  - If experiment 1 can result in any one of *m* possible outcomes and if, for each outcome of experiment 1, there are *n* possible outcomes of experiment 2,

    Then together there are *mxn* possible outcomes of the two experiments.

Example 2.1

- A small community consists of 10 women, each of whom has 3 children. If one woman and one of her children are to be chosen as mother and child of the year, how many different choices are possible?

Example 2.2

- A college planning committee consists of 3 freshmen, 4 sophomores, 5 juniors, and 2 seniors. A subcommittee of 4, consisting of 1 person from each class, is to be chosen. How many different subcommittees are possible?

Example 2.3

- How many different 7-place license plates are possible if the first 3 places are to be occupied by letters and the final 4 by numbers?

Example 2.4

- In Example 2.3, how many license plates would be possible if repetition among letters or numbers were prohibited?

Permutation

- Counting the number of ways to order things $\frac{n!}{r!}$
- Ex: How many different ordered arrangements of letters a, b, and c are possible?
- Suppose that there are *n* distinct objects, how many different ordered arrangements are possible?

Example 2.5

- A football team needs to arrange 5 players for the penalty kicks. How many arrangements are possible?

Example 2.6

- A class in probability theory consists of 6 men and 4 women. An examination is given, and the students are ranked according to their performance. Assume that no two students obtain the same score. How many different rankings are possible?

Example 2.7

- Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

จัดเรียงปกติ $10!$

วิชาเดียวกันติดกัน $\dfrac{10!}{(4!\,3!\,2!\,1)\,4!}$

Example 2.8

- How many different letter arrangements can be formed from the letters *PEPPER*?

Permutation of non-distinct objects

- If there are n objects, of which n_1 are alike, n_2 are alike, …, n_r are alike, then the number of permutations is

$$\frac{n!}{n_1! \cdot n_2! \dots n_r!}$$

Example 2.9

- A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil. If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

Sampling

- Sampling with replacement and with order n^k
- Sampling without replacement and with order $\frac{n!}{(n-k)!}$
- Sampling without replacement and without order $\frac{n!}{(n-k)!k!}$
- Sampling with replacement and without order $\binom{n-1+k}{k}$

$$\frac{(n-1+k)!}{(n-k)!\,k!}$$

Sampling w/ replacement w/ order

- Pick k objects from a set of n distinct objects with replacement and with order

- Ex: An urn contains 5 balls numbered 1-5. Select 2 balls from an urn with replacement. How many distinct ordered pairs are possible?

Example 2.10

- Given that 12 airplane crashes occur at random in a year, what is the probability that there is exactly 1 crash each month?

$\Omega = 12^{12}$

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$P(A) = \frac{12!}{12^{12}}$

Sampling w/o replacement w/ order

- Pick k objects from a set of n distinct objects without replacement and with order

- Ex: An urn contains 5 balls numbered 1-5. Select 2 balls from an urn without replacement. How many distinct ordered pairs are possible?

Sampling w/o replacement w/o order

- Pick k objects from a set of n distinct objects without replacement and without order

- Ex: An urn contains 5 balls numbered 1-5. Select 2 balls from an urn without replacement. How many distinct pairs are possible if we neglect the order?

Example 2.11

- A lunch set allows a person to pick 3 sides. If there are 8 sides available on the menu to choose from, how many distinct selections are possible? Assuming that no side can be chosen more than once.

$$\binom{8}{3} = \frac{8!}{5!\,3!}$$

Example 2.12

- A committee of 3 is to be formed from a group of 20 people. How many different committees are possible?

Example 2.13

- A batch of 50 items contains 10 defective items. If 10 items are chosen for testing

  $\binom{50}{10}$ $\frac{50!}{40!\,10!}$

  - How many possible ways are there to sample the items?
  - How many ways are there to get 5 good items and 5 bad items?

  $\binom{40}{5}\binom{10}{5}$

Binomial Coefficient

- $\binom{n}{k}$ is usually called a *binomial coefficient*

- Binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Multinomial Coefficient

- A set of n distinct items is to be divided into r distinct groups of respective sizes $n_1, n_2, \ldots, n_r$, where $n_1 + n_2 + \ldots + n_r = n$. How many different divisions are possible?

Example 2.14

- A police department in a small city consists of 10 officers. If the department policy is to have 5 of the officers patrolling the streets, 2 of the officers working full time at the station, and 3 of the officers on reserve at the station, how many different divisions of the 10 officers into the 3 groups are possible?

Sampling w/ replacement w/o order

- Pick k objects from a set of n distinct objects with replacement and without order

- Ex: Pick 2 ice-cream scoops out of 4 available flavors. Flavors can be repeated.